1 Dataset preparation and leakage-aware data audit

Original dataset

651,191 raw URLs
Benign 428,103 | Defacement 96,457
Phishing 94,111 | Malware 32,520

→

Integrity audit

URL deduplication and label-conflict removal
10,078 records removed

→

Fixed stratified split

641,113 retained URLs
Train 512,890 | Validation 64,111 | Test 64,112

→

Training-only preparation

Character/token vocabularies; class weights
$L_c = 200$ | $L_t = 40$; seeds 42, 123, 2025

↓

2 Dual-granularity URL representation

Input: raw URL u_i (character and token views created independently)

Character view — fine-grained morphology

Character sequence -> embedding -> parallel Conv1D (k = 3, 5, 7) -> global max pooling

Token view — higher-level lexical context

Token sequence -> embedding + positional context -> masked self-attention -> masked pooling

↓

3 URLFusionNet: stabilized adaptive character-token fusion

Common projection

Character z^c | Token z^t
Shared latent dimension + LayerNorm

→

Core novelty — stabilized adaptive gate

$s = [z^c \,||\, z^t]$ -> GELU MLP -> $\mathrm{softmax}(\alpha^c, \alpha^t)$
$f = \alpha^c z^c + \alpha^t z^t$; $\alpha^c + \alpha^t = 1$
Balanced initial weights (0.5, 0.5); KL penalty $\lambda = 0.005$ (training only)

→

Residual refinement

Residual MLP
+ LayerNorm

→

Four-class output

Benign | Defacement
Phishing | Malware

No explicit Hadamard-product or absolute-difference interaction in the proposed gate.

↓

4 Experimental evaluation, component evidence and limitations

Fusion comparison

Original adaptive vs fixed averaging
vs stabilized adaptive; 3 seeds

Multiclass metrics

Macro-F1 (primary); accuracy, MCC
Class-wise F1 and confusion matrices

Gate diagnostics

Per-URL character/token weights
Entropy and collapse detection

Historical component tests

Single-view ablation and baselines
Archived perturbation tests

Efficiency and scope

Archived latency/model footprint
External generalization not tested

Historical ablation, robustness and efficiency belong to the earlier adaptive model, not the final KL-stabilized checkpoint.

Fig. 1. Research framework of the URLFusionNet study.